# Lecture 5: Quantum Information Science for Quantum Chemistry

## 1. Why Bother

Every method in Lectures 1–4 was an attempt to escape the same wall. Full CI, the exact answer in a given basis, requires $\binom{M}{N_\uparrow}\binom{M}{N_\downarrow}$ amplitudes: exponential in system size. Everything we did (mean fields, functionals, truncated excitation hierarchies, active spaces) was a compression scheme, and each compression failed somewhere, generally at strong correlation.

Feynman's 1982 observation was that the exponential is an artifact of simulating quantum mechanics with classical bits. A register of $M$ qubits has a Hilbert space of dimension $2^M$: it stores a state vector in the full Hilbert space *natively*, with one qubit per spin orbital. The question is whether we can (i) write the molecular Hamiltonian in terms of qubit operators, (ii) prepare a state with substantial overlap with the ground state, and (iii) extract the energy. This lecture covers all three, with emphasis on the two algorithmic ingredients that do the real work: the Jordan–Wigner transformation and quantum phase estimation.

## 2. Qubits, Gates, Measurement

A qubit is a two-level system, $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$ with $|c_0|^2 + |c_1|^2 = 1$. For $M$ qubits, $|\psi\rangle = \sum_{\mathbf{n}\in\{0,1\}^M} c_{\mathbf{n}}|\mathbf{n}\rangle$, i.e. $2^M$ complex amplitudes. The single-qubit Pauli operators are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{1}$$

so $X|0\rangle = |1\rangle$ (a bit flip) and $Z|n\rangle = (-1)^n|n\rangle$ (a phase flip). Two more ingredients we need. The **Hadamard** gate creates superposition,

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \equiv |+\rangle, \qquad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \equiv |-\rangle, \qquad H^2 = I. \tag{2}$$

And any unitary $V$ can be made **controlled** on a qubit $c$: the gate c-$V$ acts as the identity when $c$ is $|0\rangle$ and applies $V$ to the target when $c$ is $|1\rangle$,

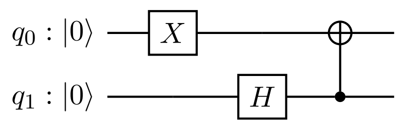
$$\text{c-}V\ \left(\alpha|0\rangle + \beta|1\rangle\right)|\chi\rangle = \alpha|0\rangle|\chi\rangle + \beta|1\rangle\, V|\chi\rangle. \tag{3}$$

The controlled-NOT, $\text{CNOT}_{c\to t}$, which flips the target iff the control is $|1\rangle$, is the special case $V = X$, and it is the standard way to create entanglement. Equation (3) is worth staring at: the control qubit ends up in a superposition of "$V$ was applied" and "$V$ was not applied". That is the entire mechanism behind phase estimation in Section 6.

All operations are unitary and therefore reversible; the only irreversible step is measurement, which returns a bit string $\mathbf{n}$ with probability $|c_{\mathbf{n}}|^2$ and destroys the superposition. This is the central practical constraint: we can store $2^M$ amplitudes but we can only read out $M$ bits per shot. Useful algorithms are those that arrange interference so that the bits we read are the ones we want.

**A three-gate example.** Start from $|00\rangle$; apply $X$ to $q_0$, $H$ to $q_1$, then $\mathrm{CNOT}_{1\to 0}$:

$$|00\rangle \xrightarrow{X_0} |10\rangle \xrightarrow{H_1} \frac{1}{\sqrt{2}}\left(|10\rangle + |11\rangle\right) \xrightarrow{\mathrm{CNOT}_{1\to 0}} \frac{1}{\sqrt{2}}\left(|10\rangle + |01\rangle\right). \tag{4}$$




Arrange the amplitudes into the matrix $C_{q_0q_1} = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$: it has rank 2, so the state is not a product: it is entangled. And, as we are about to see, it is an eigenstate of the number operator with exactly one particle. Three gates have produced the smallest possible correlated wavefunction.

## 3. From Fock Space to Qubits

Second quantization (Lecture 3) already gave us the right representation. A determinant is an occupation string; assign one qubit per spin orbital,

$$|n_0 n_1 \cdots n_{M-1}\rangle, \qquad n_p \in \{0, 1\}, \tag{5}$$

with spin orbital 0 in the leftmost bit. The occupation and total-number operators are diagonal:

$$\hat{n}_p = \frac{I - Z_p}{2}, \qquad \hat{N} = \sum_{p=0}^{M-1} \hat{n}_p. \tag{6}$$

Acting on any occupation string, $Z_p|\mathbf{n}\rangle = (-1)^{n_p}|\mathbf{n}\rangle$, so $\hat{n}_p|\mathbf{n}\rangle = n_p|\mathbf{n}\rangle$ and

$$\hat{N}|\mathbf{n}\rangle = \left(\sum_p n_p\right)|\mathbf{n}\rangle : \tag{7}$$

every computational basis state is a particle-number eigenstate. Minimal-basis $H_2$ in the RHF determinant is $|1100\rangle$ with $\hat{N} = 2$, using the ordering $(\sigma_g\alpha, \sigma_g\beta, \sigma_u\alpha, \sigma_u\beta)$.

A general superposition is *not* a number eigenstate, and this matters. If our ansatz can leak amplitude into the $N \pm 1$ sectors, the variational energy is minimized over the wrong space and can fall below the true $N$-electron ground state. Ansatze built from particle-number-conserving operators, namely excitations $a_a^\dagger a_i$ that move an electron rather than creating one, stay in the correct symmetry sector by construction. The same argument applies to $S_z$, $S^2$, and point group symmetry, and enforcing them is both a correctness and an efficiency measure.

## 4. The Jordan–Wigner Transformation

Occupations map onto bits trivially. The subtlety is antisymmetry: qubit operators on different sites commute, while fermionic operators anticommute. Jordan and Wigner (1928) resolved this by attaching a string of $Z$'s that counts the parity of all preceding orbitals:

$$a_p^\dagger = \left(\prod_{q<p} Z_q\right)\frac{X_p - iY_p}{2}, \qquad a_p = \left(\prod_{q<p} Z_q\right)\frac{X_p + iY_p}{2}. \tag{8}$$

The local part is a raising/lowering operator: $(X - iY)/2 = |1\rangle\langle 0|$ and $(X + iY)/2 = |0\rangle\langle 1|$. The $Z$ string supplies the $(-1)^{\sum_{q<p} n_q}$ sign that a determinant picks up when an electron is created past occupied orbitals; that sign is exactly what makes $\{a_p, a_q^\dagger\} = \delta_{pq}$ hold.

**Worked example: two modes.** With $a_0 = \frac{1}{2}(X_0 + iY_0)$ and $a_1 = \frac{1}{2}Z_0(X_1 + iY_1)$, use $(X - iY)Z = X - iY$ and $Z(X + iY) = X + iY$ (both follow from $Z|0\rangle\langle 1| = |0\rangle\langle 1|$) to eliminate the strings:

$$a_0^\dagger a_1 = \tfrac{1}{4}(X_0 - iY_0)Z_0(X_1 + iY_1) = \tfrac{1}{4}(X_0 - iY_0)(X_1 + iY_1), \tag{9}$$

$$a_1^\dagger a_0 = \tfrac{1}{4}Z_0(X_1 - iY_1)(X_0 + iY_0) = \tfrac{1}{4}(X_0 + iY_0)(X_1 - iY_1). \tag{10}$$

Adding, the cross terms cancel and

$$\boxed{a_0^\dagger a_1 + a_1^\dagger a_0 = \tfrac{1}{2}\left(X_0X_1 + Y_0Y_1\right).} \tag{11}$$

This is the hopping (one-body off-diagonal) term, the fundamental building block of any electronic Hamiltonian. It manifestly conserves particle number: acting on $|10\rangle$ it gives $|01\rangle$ and vice versa, while annihilating $|00\rangle$ and $|11\rangle$. It moves one electron between two modes rather than changing their total. Equivalently, $[\hat{N}, a_0^\dagger a_1 + a_1^\dagger a_0] = 0$.

Applying (8) to the second-quantized Hamiltonian

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2}\sum_{pqrs}\langle pq|rs\rangle a_p^\dagger a_q^\dagger a_s a_r \;\longrightarrow\; \hat{H} = \sum_j c_j\,\hat{P}_j, \qquad \hat{P}_j \in \{I, X, Y, Z\}^{\otimes M}, \tag{12}$$

gives a sum of $\mathcal{O}(M^4)$ Pauli strings with real coefficients $c_j$ built from the molecular integrals. Every ingredient is a quantity you already know how to compute with a classical quantum chemistry code. Two practical notes: JW strings have weight up to $\mathcal{O}(M)$, which costs gates; the Bravyi–Kitaev and parity encodings reduce this to $\mathcal{O}(\log M)$. And known symmetries let you remove qubits entirely, and minimal-basis $H_2$ reduces from four qubits to one.

One consequence deserves a moment. Two modes in the one-particle sector admit exactly two Fock states, $|10\rangle$ and $|01\rangle$, and the register holds an equal-weight superposition of them — maximal static correlation, the configuration that defeated every single-reference method of Lectures 1–4 — at exactly the cost of holding a single determinant. The difficulty that organized four lectures is an artifact of expanding in determinants, not a property of the state. That is the promise of this lecture. The rest of it is about the harder problem: getting a number back out.

## 5. Energies Are Phases: the One-Qubit Circuit

One strategy is available immediately: prepare a parameterized state $|\psi(\boldsymbol{\theta})\rangle$, measure $E(\boldsymbol{\theta}) = \sum_j c_j\langle\psi(\boldsymbol{\theta})|\hat{P}_j|\psi(\boldsymbol{\theta})\rangle$ term by term, and hand the numbers to a classical optimizer — the variational principle of Lecture 1 with the trial function held in quantum rather than classical memory. It does not scale, and the obstruction is measurement rather than expressibility: each Pauli expectation needs $\mathcal{O}(1/\varepsilon^2)$ samples, there are $\mathcal{O}(M^4)$ of them, and chemical accuracy then demands an absurd number of circuit repetitions. Grouping commuting terms and better estimators shave the prefactor, never the $1/\varepsilon^2$.

So change strategy completely. Rather than averaging an observable we will *measure* the energy, exploiting the fact that on a quantum computer an eigenvalue is naturally a phase.

Let $U(t) = e^{-i\hat{H}t}$ for some fixed $t > 0$. For an exact eigenstate,

$$U(t)|\Phi_k\rangle = e^{-iE_k t}|\Phi_k\rangle \equiv e^{2\pi i\phi_k}|\Phi_k\rangle, \qquad \phi_k \in [0,1). \tag{13}$$

The second equality is only a change of variable, and it is worth inverting right away so that $\phi_k$ does not feel like a detour. Matching exponents, $2\pi\phi_k = -E_k t \pmod{2\pi}$, hence

$$E_k = -\frac{2\pi\phi_k}{t} + \frac{2\pi n}{t}, \qquad n \in \mathbb{Z}. \tag{14}$$

The circuit supplies $\phi_k$, we choose $t$, and the integer $n$ has to come from knowing $E_k$ in advance to within $2\pi/t$ — in practice by shifting the Hamiltonian by a classical estimate, $\hat{H} \to \hat{H} - E_{\text{est}}$, so the residual eigenvalue lands inside a single window and $n = 0$. So $\phi_k \in [0,1)$ describes the phase completely but the energy only modulo $2\pi/t$.

Note that $U$ does nothing to an eigenstate except multiply it by a number of modulus one. Measuring the state itself therefore tells us nothing: global phases are unobservable. The trick is to move that phase onto a *separate* qubit, where it becomes a relative phase and hence measurable.

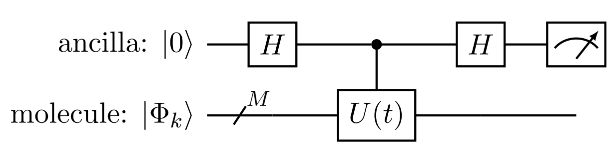
Take one extra qubit, an **ancilla**: a work qubit that carries no chemistry, in addition to the $M$ qubits holding the wavefunction. Prepare the ancilla in $|+\rangle$ with a Hadamard and apply the controlled evolution (3) with $V = U(t)$:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}|\Phi_k\rangle \xrightarrow{\text{c-}U} \frac{|0\rangle|\Phi_k\rangle + |1\rangle\, U|\Phi_k\rangle}{\sqrt{2}} = \left(\frac{|0\rangle + e^{2\pi i\phi_k}|1\rangle}{\sqrt{2}}\right)|\Phi_k\rangle. \tag{15}$$

Read the right-hand side carefully. The molecular register comes out *unchanged*, still in $|\Phi_k\rangle$; the eigenvalue has been transferred to the ancilla as a relative phase between $|0\rangle$ and $|1\rangle$. This is called **phase kickback**, and it is the whole idea. Because the eigenstate survives intact, we may do it again and again.

To read the phase out, undo the Hadamard and measure the ancilla in the computational basis:

$$H\left(\frac{|0\rangle + e^{2\pi i\phi}|1\rangle}{\sqrt{2}}\right) = \frac{1 + e^{2\pi i\phi}}{2}|0\rangle + \frac{1 - e^{2\pi i\phi}}{2}|1\rangle, \qquad p(0) = \cos^2\pi\phi, \quad p(1) = \sin^2\pi\phi. \tag{16}$$




This circuit is the *Hadamard test*. It returns one *classical* bit per shot — the ancilla is a single qubit, so each repetition yields a 0 or a 1, not a digit of $\phi$. Assigning $\pm 1$ to those two outcomes, the quantity we actually estimate is the expectation value of $Z$ on the ancilla,

$$\langle Z\rangle = p(0) - p(1) = \cos^2\pi\phi - \sin^2\pi\phi = \cos 2\pi\phi, \tag{17}$$

so the phase shows up at twice the angle: $\cos^2\pi\phi$ is a probability, whereas $\cos 2\pi\phi$ is the estimator built from the difference of the two probabilities. This, however, is not enough to pin $\phi$ down.

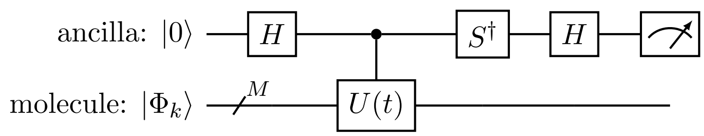
Cosine is even, so $\cos 2\pi\phi = \cos 2\pi(1-\phi)$ and this circuit produces *identical* statistics for $\phi$ and $1-\phi$ — the sign of the residual energy inside the $2\pi/t$ window drops out. Nothing is wrong with the test; we have simply measured only the real part of $e^{2\pi i\phi}$, and locating a point on the unit circle takes both coordinates.

The imaginary part comes from an almost identical circuit. Insert the phase gate $S^\dagger = \mathrm{diag}(1, -i)$ on the ancilla between the controlled evolution and the final Hadamard. It rotates the kicked-back phase back by a quarter turn,

$$\frac{|0\rangle + e^{2\pi i\phi}|1\rangle}{\sqrt{2}} \xrightarrow{S^\dagger} \frac{|0\rangle + e^{i(2\pi\phi - \pi/2)}|1\rangle}{\sqrt{2}}, \tag{18}$$

after which the algebra above goes through verbatim with $2\pi\phi$ replaced by $2\pi\phi - \pi/2$:

$$p(0) = \cos^2\!\left(\pi\phi - \tfrac{\pi}{4}\right), \quad p(1) = \sin^2\!\left(\pi\phi - \tfrac{\pi}{4}\right), \qquad \langle Z\rangle = \cos\!\left(2\pi\phi - \tfrac{\pi}{2}\right) = \sin 2\pi\phi. \tag{19}$$




The two circuits together estimate the real and imaginary parts of $e^{2\pi i\phi}$, which determine $2\pi\phi = \arg\left(\cos 2\pi\phi + i \sin 2\pi\phi\right)$ uniquely on $[0, 2\pi)$.

In either version we are averaging a Bernoulli variable, so the error falls off only as $1/\sqrt{\text{shots}}$ and we are back to $\mathcal{O}(1/\varepsilon^2)$ cost. The next section fixes precisely this.

## 6. Quantum Phase Estimation

The improvement comes from two observations.

**Repetition sharpens the phase.** Applying $U$ twice doubles the phase: $U^2|\Phi_k\rangle = e^{2\pi i(2\phi_k)}|\Phi_k\rangle$. In general $U^{2^j}$ kicks back $e^{2\pi i 2^j \phi_k}$. Write $\phi_k$ as a binary fraction,

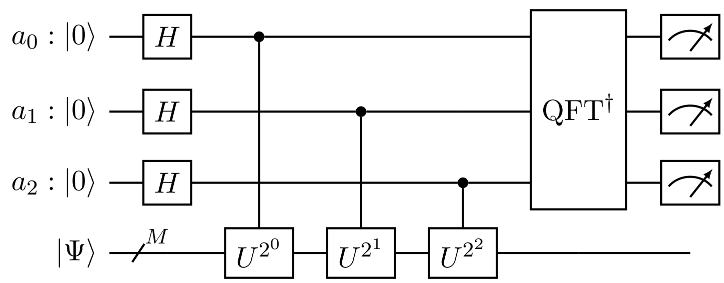
$$\phi_k = 0.y_1y_2y_3\ldots = \frac{y_1}{2} + \frac{y_2}{4} + \cdots, \qquad y_i \in \{0, 1\}. \tag{20}$$

Multiplying by $2^j$ shifts the binary point $j$ places right; the bits that move to the left of the point are integers, and $e^{2\pi i\,(\text{integer})} = 1$, so they are invisible. Each power of $U$ therefore exposes a different, deeper bit of $\phi_k$. Small phases are hard to distinguish from zero in one shot, but the same phase amplified $2^j$ times is not.

**One ancilla per bit.** Use $m$ ancillas, all prepared in $|+\rangle$, and let ancilla $j$ ($j = 0, \ldots, m-1$) control $U^{2^j}$. Applying (15) to each in turn, the ancilla register ends up in

$$\frac{1}{\sqrt{2^m}} \bigotimes_{j=0}^{m-1} \left(|0\rangle + e^{2\pi i 2^j \phi_k}|1\rangle\right) = \frac{1}{\sqrt{2^m}} \sum_{x=0}^{2^m-1} e^{2\pi i \phi_k x}|x\rangle, \tag{21}$$

where expanding the product and reading the bits of $x$ as the choice of $|0\rangle$ or $|1\rangle$ in each factor gives the sum. Every basis state of the register now carries the phase, wound at a different rate. The full circuit, drawn for $m = 3$:




Read it left to right: Hadamards make the ancillas into $|+\rangle$, each controlled $U^{2^j}$ kicks its amplified phase onto ancilla $j$, and the inverse Fourier transform converts those relative phases into the bit string we measure. The molecular register at the bottom is never measured, and comes out unchanged.

Equation (21) is exactly a discrete Fourier transform. If $\phi_k = y/2^m$ for an integer $y \in [0, 2^m - 1]$, i.e. if $\phi_k$ has an exact $m$-bit binary expansion, then

$$\frac{1}{\sqrt{2^m}} \sum_x e^{2\pi i x y / 2^m} |x\rangle \;\equiv\; \mathrm{QFT}|y\rangle, \tag{22}$$

the **quantum Fourier transform** of the computational basis state $|y\rangle$. The QFT is a unitary on $m$ qubits implementable with $\mathcal{O}(m^2)$ elementary gates, so we can simply apply its inverse:

$$\mathrm{QFT}^\dagger \left( \frac{1}{\sqrt{2^m}} \sum_x e^{2\pi i x y / 2^m} |x\rangle \right) = |y\rangle. \tag{23}$$

Using $\mathrm{QFT}^\dagger|x\rangle = 2^{-m/2} \sum_s e^{-2\pi i x s / 2^m} |s\rangle$, the amplitude left on $|s\rangle$ after the inverse transform is a geometric sum,

$$\frac{1}{2^m} \sum_{x=0}^{2^m - 1} e^{2\pi i x (\phi_k - s/2^m)} = \delta_{sy} \qquad \text{when } \phi_k = y/2^m, \tag{24}$$

where all $2^m$ amplitudes return zero except one (i.e., $s = y$). That is exactly what the Hadamard test could not do: there the ancilla was left in a superposition, so its measured bit was random and had to be averaged, whereas here the inverse Fourier transform collapses the phase information onto a single basis state. Measuring the register therefore returns the $m$ bits of $y$ *deterministically* in one shot — and those $m$ bits are precisely the binary digits $y_1 \dots y_m$ of $\phi_k$ from (20), one per ancilla. Mind the ordering: in the convention of (21), where ancilla $j$ carries the $2^j$ place of the integer, the bit measured on ancilla $j$ is $y_{m-j}$, so the register reads out the digits of $\phi_k$ backwards.

Lastlt, from (13) with $\phi_k = y/2^m$,

$$-E_k t = 2\pi \phi_k \pmod{2\pi} \quad \Longrightarrow \quad E_k = -\frac{2\pi y}{t\, 2^m} + \frac{2\pi n}{t}, \quad n \in \mathbb{Z}. \tag{25}$$

The integer $n$ is a genuine *aliasing* branch: a phase is only defined modulo $2\pi$, so the measurement determines $E_k$ only within a window of width $2\pi/t$. $t$ **cannot be made arbitrarily large**, but we can always add more ancillae. The resolution follows from $y$ being an *integer*, so that the possible answers form a grid: stepping $y \to y + 1$ above moves the energy by

$$\boxed{\delta E \simeq \frac{2\pi}{t\, 2^m}.} \tag{26}$$

Equivalently, and more memorably, $\delta E$ is the window $2\pi/t$ cut into the $2^m$ bins that $m$ bits are able to label. Two eigenvalues closer together than $\delta E$ land on the same readout and cannot be told apart, which is why it is $\delta E$, not the window, that has to be compared against chemical accuracy. Demanding $\delta E \leq \varepsilon_E$ where $\varepsilon_E$ is the target accuracy gives

$$2^m \geq \frac{2\pi}{t\,\varepsilon_E} \qquad \Longrightarrow \qquad m \geq \left\lceil \log_2 \frac{2\pi}{t\,\varepsilon_E} \right\rceil. \tag{27}$$

This exposes the central tradeoff. Increasing $t$ improves resolution at fixed $m$, but shrinks the unambiguous window $2\pi/t$, so you need prior knowledge of $E_k$ to within that window (in practice from a classical calculation) to fix $n$. Increasing $m$ instead keeps the window but adds ancillas and, more importantly, requires implementing $U(t)^{2^{m-1}}$: coherent evolution exponentially longer than a single application.

The circuit applies $U$ a total of

$$\sum_{j=0}^{m-1} 2^j = 2^m - 1 \tag{28}$$

times, so the total evolution time is $T = t\,(2^m - 1) \approx t\,2^m = 2\pi/\delta E$. Getting one more bit of the energy means doubling how long the device must stay coherent. But note what this buys:

$$\underbrace{T \sim \frac{1}{\varepsilon_E}}_{\text{QPE: coherent evolution, one run}} \qquad \text{versus} \qquad \underbrace{N_{\text{shots}} \sim \frac{1}{\varepsilon_E^2}}_{\text{sampling: independent short runs}} \quad . \tag{29}$$

The price for the Heisenberg scaling is that all of that evolution must be coherent, which is what makes QPE a fault-tolerant algorithm rather than a near-term one.

**Two caveats.** When $\phi_k$ is not an exact $m$-bit fraction the transform concentrates on the nearest $y$ instead, with probability at least $4/\pi^2 \approx 0.41$, and on the two bins bracketing $\phi_k$ with probability $\approx 0.81$; the rest decays as $1/k^2$ in the number of bins $k$ away, so a miss is a mis-rounding by a few $\delta E$ rather than a wrong eigenvalue. Getting the leading $m$ bits right with probability $1 - \eta$ costs only $m + \mathcal{O}(\log 1/\eta)$ ancillas, so we take the ideal case. The discussion above assumed the molecular register really is in an eigenstate $|\Phi_k\rangle$ — Section 7 handles what happens when it is not. Remember that for most applications we do not know the eigenstate in advance, so we have to prepare an approximate eigenstate!

## 7. The State Preparation Problem

Everything above assumed the molecular register was already in an eigenstate. But if we had one, we would be done. In practice we load a classically computed trial state $|\Psi\rangle = \sum_k d_k |\Phi_k\rangle$. The circuit is linear, so it acts on each eigencomponent independently and the two registers come out *entangled*:

$$\sum_k d_k |y_k\rangle_{\text{ancilla}} |\Phi_k\rangle_{\text{molecule}}. \tag{30}$$

Measuring the ancillas gives the bit string $y_k$, and hence the energy $E_k$, with probability

$$p_k = |d_k|^2 = |\langle \Phi_k | \Psi \rangle|^2, \tag{31}$$

collapsing the molecular register onto the corresponding eigenstate. So QPE is a projective measurement of the energy: it returns an *exact* eigenvalue (to the resolution $\delta E$), and your trial state controls only *which* one. Expect $\mathcal{O}(1/p_0)$ repetitions to land on the ground state. Usefully, the collapsed state is then a genuine eigenstate, which can be reused.

Collecting the pieces, the cost of a ground-state energy is

$$\boxed{C \;\sim\; \frac{1}{\gamma^2}\left(C_{\text{prep}} + \frac{1}{\varepsilon_E}\right), \qquad \gamma = |\langle\Phi_0|\Psi\rangle|} \tag{32}$$

with $1/\gamma^2 = 1/p_0$ the expected number of attempts and the bracket the price of one attempt: build the trial state, then run the phase estimation. The two terms *add* because the circuit of one run is simply the preparation followed by the phase estimation, so what adds is *depth*: a single preparation buys the whole $1/\varepsilon_E$ of evolution, since nothing re-prepares the register while it accumulates phase. Note that this depth is inside the coherent window, so $C_{\text{prep}}$ is charged against the coherence budget like everything else — it is simply small, since $C_{\text{prep}} \ll 1/\varepsilon_E$ at chemical accuracy. State preparation is therefore cheap *per attempt* and does its damage through $\gamma$ in the prefactor. Contrast the sampling strategy of Section 5, where every one of the $1/\varepsilon_E^2$ shots needs its own fresh copy of $|\Psi\rangle$, so there $C_{\text{prep}}$ multiplies instead. Amplitude amplification improves the prefactor from $1/\gamma^2$ to $1/\gamma$, and the best known ground-state energy algorithms achieve $\widetilde{\mathcal{O}}(1/(\gamma\,\varepsilon_E))$ overall.

$\gamma$ is where the biggest (subtle) danger is. For a trial state that is a product of $n$ locally accurate fragments each with overlap $s < 1$, the total overlap, $\gamma$, decays as $s^n$, exponentially in system size. Whether chemically relevant systems have efficiently preparable states of substantial overlap is an open question, and current evidence does not support a generic exponential speedup for ground-state chemistry. The plausible claims are narrower: specific strongly correlated systems where classical methods are unreliable and a good active-space state is preparable.

## 8. Implementing the Time Evolution

Every cost so far has been quoted in units of *evolution*: the coherent time $T = 2\pi/\delta E$, or equivalently the $2^m - 1$ applications of $U(t)$ that make it up. Neither of those is a gate count. What is still missing is

$$\#\,\text{gates} \;=\; \underbrace{T}_{\text{Sections 6–7}} \;\times\; \underbrace{\text{gates per unit of evolution}}_{\text{this section}}, \tag{33}$$

and we have so far said nothing at all about the second factor. We still must realize $U(t) = e^{-i\hat{H}t}$ from a Hamiltonian $\hat{H} = \sum_j c_j \hat{P}_j$ whose terms do not commute.

**Trotter methods.** The textbook route is Trotterization,

$$e^{-i\hat{H}t} = \left(\prod_j e^{-ic_j\hat{P}_j t/n}\right)^n + \mathcal{O}\left(\frac{t^2}{n}\sum_{j<k}\|[\hat{P}_j, \hat{P}_k]\|\right), \tag{34}$$

where each factor $e^{-ic_j\hat{P}_j\tau}$ is a short, exactly implementable circuit (basis change, CNOT ladder, $Z$-rotation, uncompute). It is quite hard to obtain a tight bound on the Trotter error in practice, and the best known bounds are often far too pessimistic. The most useful data points are usually based on simulations over specific chemical examples.

Modern algorithms — qubitization and quantum signal processing — take a different route, and achieve cost linear in $t$ and logarithmic in $1/\varepsilon$.

**Block encoding, LCU, and qubitization.** $\hat{H}$ is not unitary, so no circuit applies it directly; the trick is to hide it inside one that is. A unitary $U_{\hat{H}}$ acting on $a$ ancillas plus the system *block encodes* $\hat{H}$ when

$$\left(\langle 0|^{\otimes a} \otimes I\right) U_{\hat{H}} \left(|0\rangle^{\otimes a} \otimes I\right) = \frac{\hat{H}}{\lambda}, \tag{35}$$

i.e. when $\hat{H}/\lambda$ occupies one block of it. The standard construction is the *linear combination of unitaries* (LCU): since Jordan–Wigner already wrote $\hat{H} = \sum_j c_j \hat{P}_j$ as a sum of unitaries, take $U_{\hat{H}} = \mathrm{PREP}^{\dagger}\,\mathrm{SEL}\,\mathrm{PREP}$ with

$$\mathrm{PREP}|0\rangle = \sum_j \sqrt{c_j/\lambda}\,|j\rangle, \qquad \mathrm{SEL}|j\rangle|\psi\rangle = |j\rangle\hat{P}_j|\psi\rangle, \tag{36}$$

and the normalization this forces is exactly the $\ell_1$ norm $\lambda = \sum_j |c_j|$. So $\lambda$ is not an artifact of anyone's error analysis: it is the subnormalization of the encoding, which is why every cost in this literature is quoted in units of it.

Qubitization is then what one *does* with such an encoding — reflect about the ancilla $|0\rangle$ subspace after each use of $U_{\hat{H}}$ and the resulting walk operator has eigenphases $\arccos(E_k/\lambda)$, so the spectrum can be phase-estimated without simulating time at all. Read that literally: there is no $t$ anywhere and $e^{-i\hat{H}t}$ is never built. We phase-estimate a fixed unitary that is not a propagator, inverting $E_k = \lambda\cos\theta_k$ in place of (14). Two consequences follow. There is no Trotter error, since nothing has been discretized. And because $\lambda \geq \|\hat{H}\|$ by construction, $E_k/\lambda$ lies in $[-1, 1]$ where arccos is single-valued, so the aliasing branch $n$ of Section 6 never arises at all. The phase is of course still defined only modulo $2\pi$; what has changed is that arccos cannot overflow a period, whereas $E_k t$ can. This is the "choose $t$ small enough that the window covers the spectral range" escape of Section 6, now enforced automatically by $\lambda$. The only structure left is a harmless twofold degeneracy: $W$ has eigenvalues $e^{\pm i\theta_k}$, so the readout returns $\theta_k$ or $2\pi - \theta_k$, and cos is even. Note what has *not* changed: qubitization replaces the unitary being phase-estimated, not the phase estimation. The Hadamards, the controlled powers and the inverse QFT of Section 6 are all still there, applied to $W$ instead of to $U(t)$. Which unitary encodes the spectrum and how one reads a phase out of it are independent choices, which is why the QFT-free variants mentioned below apply equally well to either.

Note also that $\lambda$ appears differently in Trotterization and qubitization. Trotterization pays the cost due to $\lambda$ in gates per unit of evolution time, whereas qubitization pays it in the query count itself: $\mathrm{d}E/\mathrm{d}\theta = -\lambda\sin\theta$, so pinning $E_k$ to $\varepsilon$ near the middle of the spectrum requires $\theta_k$ to $\mathcal{O}(\varepsilon/\lambda)$, hence $\mathcal{O}(\lambda/\varepsilon)$ queries.

**The hunt for smaller $\lambda$.** Because every qubitized cost is quoted in units of $\lambda$, the most productive line of work has simply been to shrink it, by factorizing the two-electron integrals better: Cholesky and density fitting, then double factorization, then tensor hypercontraction, each buying a smaller $\lambda$ and a cheaper block encoding. This is the same tensor structure that gave us density fitting in Lecture 3, exploited for a different end — there it bought a prefactor on $\mathcal{O}(N^5)$, here it buys orders of magnitude of Toffoli count.

**Trotter is not obsolete.** The $\lambda$ qubitization pays is a worst-case quantity. Trotter error bounded through the commutators $[\hat{P}_j, \hat{P}_k]$ that actually appear can be far smaller than a norm bound suggests, and for some Hamiltonians Trotterized phase estimation is the cheaper option. Which approach wins is system-dependent and genuinely unsettled.

**Early fault tolerance.** Phase estimation as we derived it is a fault-tolerant algorithm: it wants $2\pi/\varepsilon_E$ of *coherent* evolution plus $m$ ancillas and a coherent QFT. Much recent work asks what is reachable with far shallower circuits, trading coherent depth for repetitions — statistical and Bayesian phase estimation, and QSVT-style eigenstate filtering, which projects onto the ground state using a polynomial approximating a step function and needs no QFT at all. These are deliberately QFT-free — a single ancilla and classical post-processing in place of the coherent transform — and the filtering variants dispense with phase estimation altogether: worse asymptotics, much better fit to the devices that will exist first.

Resource estimates for a genuinely hard target (e.g. the FeMo cofactor of nitrogenase in a large active space) currently sit in the range of millions of physical qubits and days of runtime with error correction. Far from today's hardware, but a fixed, well-defined target rather than an unknown.

## 9. Summary and Connections

- Qubit registers store FCI vectors natively; the exponential is in the *classical* representation, not in nature.
- The Jordan–Wigner transformation maps fermions to qubits, with $Z$ strings enforcing antisymmetry, and converts $\hat{H}$ into $\mathcal{O}(M^4)$ Pauli strings whose coefficients are ordinary molecular integrals.
- Measuring $\langle \hat{H} \rangle$ term by term and minimizing is the obvious strategy (i.e., VQE) but would still be prohibitively expensive: $\mathcal{O}(M^4)$ Pauli terms at $\mathcal{O}(1/\varepsilon^2)$ samples each.
- QPE measures energy projectively at $\mathcal{O}(1/\varepsilon)$ cost, at the price of long coherent evolution and a state-preparation overlap that we do not know how to guarantee.
- The bottleneck is the same one as in Lectures 1–4: finding a good approximation to the ground state. Quantum computers change what you can do *with* such a state, not the difficulty of guessing it.

Which brings the five lectures full circle. We started with a $3N$-dimensional differential equation, compressed it to a mean field, and spent three lectures repairing the compression. The quantum-computing program is the proposal to stop compressing, and its hard part turns out to be, once again, the initial guess.